

Graduation Project Proposal

Title	Recommender systems using generative retrieval based on language models
Summary (in 150 words)	We propose an intent-aware recommendation framework that dynamically re-weights item similarity based on purchase completion events. Our exploratory data analysis on the Amazon Clothing, Shoes and Jewelry dataset (4.5M interactions, 1.1M users) reveals that 84.8% of users switch categories after a purchase, yet brand loyalty persists at 26× above random chance during these transitions. Current recommender systems ignore this post-purchase behavioral shift, naively suggesting same-category items. We address this gap by constructing a Knowledge Graph (KG) that encodes User–Item–Brand–Category relationships and leveraging Large Language Models (LLMs) to dynamically infer user intent—whether brand-driven or category-driven—after each purchase event. Our approach aims to improve cross-category recommendation accuracy while maintaining explainability through faithful KG path reasoning.
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1. Project background and goals

1.1 Background

Traditional recommender systems predominantly rely on collaborative filtering (CF) or content-based filtering within a single behavioral dimension. When a user purchases a pair of Adidas running shoes, existing systems typically recommend more running shoes or similar footwear. However, real-world user behavior is far more nuanced: after completing a purchase, users often shift their interest to different categories while retaining preferences for certain attributes such as brand, style, or price range.

This phenomenon—**post-purchase category transition with attribute retention**—is well-documented in marketing literature but has received insufficient attention in the recommender systems community. The emergence of Knowledge Graphs (KGs) and Large Language Models (LLMs) provides new opportunities to model these complex, multi-dimensional user preferences.

1.2 Motivation

Our motivation stems from three key observations:

1. **Static Knowledge Graphs are insufficient:** Different users may be attracted to different aspects of a product—some prefer the brand, others the design or functionality. A static KG with fixed edge weights cannot capture this user-specific, context-dependent preference.
2. **Post-purchase intent shifts are real and measurable:** Our EDA on the Amazon Clothing, Shoes and Jewelry dataset confirms that users overwhelmingly transition to different categories after purchase (84.8%), while demonstrating statistically significant brand loyalty (26.39× above random baseline).
3. **LLMs can bridge the semantic gap:** LLMs possess the world knowledge and reasoning capability to infer why a user made a purchase and what they might want next, enabling dynamic re-weighting of KG edges based on inferred intent.

1.3 Goals

The primary goals of this project are:

- **G1** : Validate the post-purchase category transition and brand loyalty hypothesis through rigorous empirical analysis.
- **G2** : Design and implement a dynamic Knowledge Graph that adapts edge weights based on user-specific purchase intent.
- **G3** : Integrate LLM-based intent inference to dynamically guide KG traversal for cross-category recommendations.
- **G4** : Demonstrate measurable improvements over static KG baselines and traditional CF methods in cross-category recommendation scenarios.

1.4 Importance

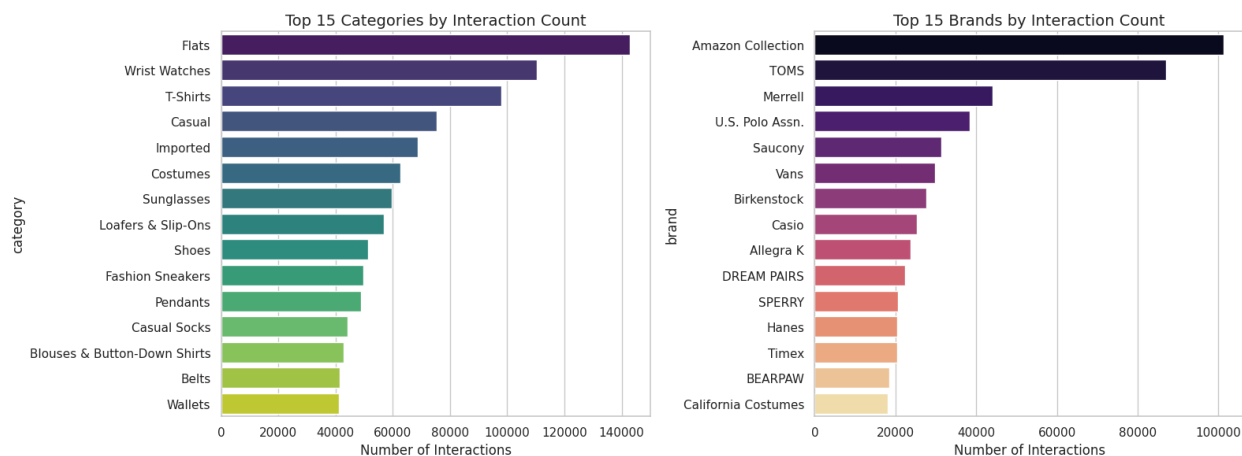
This research addresses a fundamental limitation of current recommender systems: the inability to adapt recommendations based on post-purchase behavioral shifts. By combining KG structure with LLM reasoning, we aim to create a more natural and useful recommendation experience that reflects how users actually behave after completing a purchase.

2. Project content

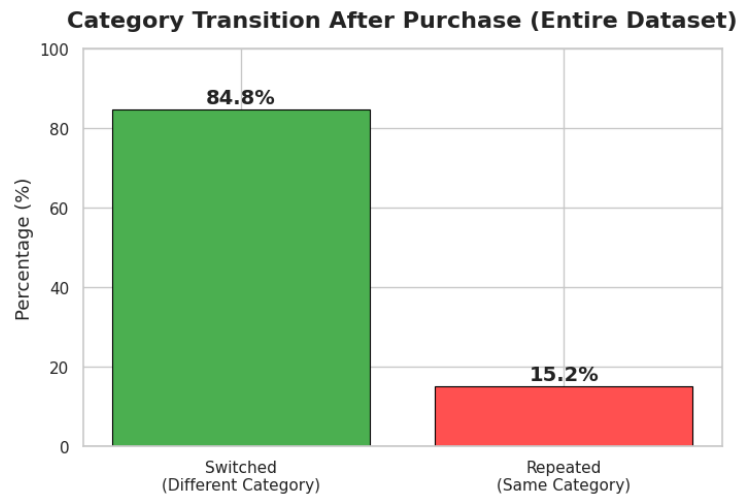
2.1 EDA Findings (Completed)

We conducted an extensive EDA on the Amazon Clothing, Shoes and Jewelry dataset (2018, 5-core) to validate our research hypothesis. The full analysis code is publicly available as a reproducible notebook¹. Note that this dataset contains only review records (ratings and text); it does not distinguish between browsing and purchasing events. We adopt the standard proxy assumption that leaving a review implies a completed purchase, which is widely used in the recommendation literature (Ni et al., 2019). All subsequent analyses—category transitions, brand retention, and repeat purchase rates—are based on this assumption. Key findings include:

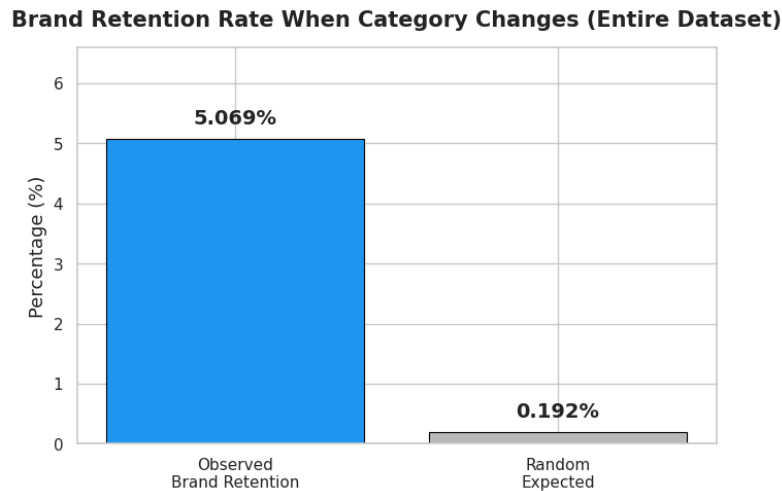
	Metric	Value
Total Interactions	4523694.0000	
Unique Users	1110563.0000	
Unique Items	160836.0000	
Unique Categories	23822.0000	
Unique Brands	21294.0000	
Avg Interactions per User	4.0700	
Sparsity (%)	99.9975	



¹ <https://colab.research.google.com/drive/1qqxGWBuxuZla9KPbeoz3Kn93pFCc-bXm?usp=sharing>



85% of users do not repurchase within the same category, strongly supporting the hypothesis that same-category recommendations post-purchase are suboptimal.



Despite switching categories, users retain brand loyalty at 26× above chance, confirming that brand serves as a bridge across categories.

Per-Category Repeat Rate

- 1,622 categories with ≥ 100 transitions were analyzed.
- The distribution of per-category repeat rates is heavily right-skewed: the median repeat rate is approximately 12%, meaning most categories exhibit low same-category repurchase behavior. However, a long tail of categories shows repeat rates above 50%, indicating that certain product types (e.g., consumables, collectibles) behave fundamentally differently from one-time-purchase items.
- This heterogeneity across categories is a critical insight: a one-size-fits-all recommendation strategy cannot accommodate the diverse post-purchase patterns observed in different product types. The model must be capable of adapting its cross-category vs. same-category recommendation ratio on a per-category (and per-user) basis.
- Data quality issue identified: Amazon metadata's hierarchical `category` field contains product attribute noise at lower levels (e.g., "Padded footbed", "Condition: Brand New" appearing as category names). Category refinement using 2nd–3rd level hierarchy is required for modeling.

2.2 From EDA Insights to Framework Design

The EDA results above establish three empirical facts: (1) users predominantly switch categories after a purchase, (2) brand loyalty persists across these transitions, and (3) transition patterns vary significantly across categories. However, these findings alone do not prescribe a specific modeling approach.

The key observation that motivates our framework is that the nature of category transitions is user-specific and context-dependent. Some users are brand-loyal shoppers who explore a brand's full product line across categories; others are style-driven explorers who seek visually similar items regardless of brand; still others exhibit purely functional transitions (e.g., buying socks after shoes). A static Knowledge Graph with fixed edge weights treats all users identically and cannot capture this heterogeneity.

This is precisely where LLM-based intent inference becomes necessary: by analyzing a user's recent interaction sequence and review text, the LLM can classify the likely transition intent (brand-driven, style-driven, function-driven, or exploratory) and dynamically re-weight the KG edges accordingly. The KG provides the structural scaffold for cross-category reasoning, while the LLM provides the personalized, context-aware signal that determines which paths to prioritize for each user at each moment.

2.3 Proposed Framework

Building on the above rationale, our framework consists of three core modules:

Module 1: Dynamic Knowledge Graph Construction

- **Entities:** User, Item, Brand, Category, Product Attributes (style, material, etc.)
- **Relations:** `purchased`, `belongs\_to\_brand`, `in\_category`, `has\_attribute`, `similar\_style`
- **Dynamic Edge Weights:** Edge weights are not static but updated based on each user's purchase history and inferred intent.
- **Category Refinement:** Use 2nd–3rd level categories from Amazon's hierarchical metadata to reduce noise identified in EDA.

Module 2: LLM-based Intent Inference

- After a purchase event, the LLM is queried with the user's recent interaction history to infer:
 - **Intent type:** brand-driven, style-driven, function-driven, or exploration
 - **Transition probability:** likelihood of staying within the same category vs. transitioning
- To enhance semantic understanding, we incorporate a GRAM-inspired multi-granular representation:
 - Item-level (specific product)
 - Attribute-level (style, material)
 - Brand-level
 - Category-level
- The LLM integrates these multi-level signals to infer:
 - dominant preference dimension
 - transition intent
- The inferred intent signals are then used to dynamically adjust the Knowledge Graph edge weights for personalized graph traversal.

Module 3: KG-guided Cross-Category Recommendation

- Step 1: Coarse Candidate Generation
 - Perform graph traversal on the dynamically weighted KG to retrieve a broad set of candidate items.
 - Paths such as `User → Brand → Other Category Items` or `User → Style → Cross-Category Items` are explored based on inferred intent.
- Step 2: LLM-guided Candidate Filtering
 - Filter the candidate set using the inferred intent signals to remove semantically irrelevant items.
- Step 3: Fine-grained Ranking (inspired by CORONA)
 - Apply a coarse-to-fine ranking strategy to progressively refine the candidate set.
 - Final ranking considers:
 - intent alignment
 - KG path scores
 - item similarity
- Faithful path reasoning provides explainability for each recommendation.

2.4 Baseline Models

We will compare our approach against the following baselines:

Model	Type	Description
BPR-MF	CF	Bayesian Personalized Ranking with Matrix Factorization
LightGCN	GNN-CF	Graph-based collaborative filtering
KGAT	KG-based	Knowledge Graph Attention Network
RippleNet	KG-based	Propagation-based KG recommendation
KGCL	KG+CL	KG-enhanced contrastive learning
K-RagRec	KG+LLM	RAG-based KG recommendation with LLM

2.5 DataSets

We will use the **RecBole** framework for standardized model implementation and evaluation.

3. Plan

3.1 [Time Line]

Semester 1

Month	Period	Goals	Details
Month 1	Mar	Prerequisites	Topic selection, key paper reading (K-RagRec, CORONA, Faithful, Boosting)
Month 2	Apr	EDA & Proposal	Dataset EDA , Proposal writing , in-depth EDA
Month 3	May	Literature Review	Related work survey, KG schema design, prior art analysis
Month 4	Jun	KG Construction	Category refinement, KG construction, baseline model training (KGAT, RippleNet, LightGCN) via RecBole

Semester 2

Month	Period	Goals
Month 5	Jul–Aug	Core model development (LLM intent inference module, dynamic edge weight mechanism)
Month 6	Sep–Oct	Integration & preliminary evaluation
Month 7	Nov–Dec	Full evaluation, ablation studies, final report & presentation

3.2

The project responsibilities are primarily divided as follows, though roles may overlap and adapt flexibly based on progress and task requirements.

Member	Primary Role	Key Responsibilities
김진욱	KG & Data Engineering	Data preprocessing, category refinement, KG construction, RecBole baseline implementation
유은선	LLM & Model Development	LLM prompt engineering for intent inference, dynamic weight module, cross-category recommendation model

Both members will collaborate closely on system integration, evaluation design, and final report writing.

4. Conclusion

4.1 Expected Technical Impact

This project addresses the underexplored problem of post-purchase behavioral shifts in recommender systems. By introducing a dynamic KG that adapts to user intent inferred via LLMs, we expect to:

- Achieve significant improvements in cross-category recommendation accuracy over static KG baselines, as measured by the proposed Cross-Category Hit Rate and Brand-Consistent Cross-Category Rate metrics.
- Provide explainable recommendations through faithful KG path reasoning, unlike black-box LLM approaches.
- Demonstrate a practical paradigm for integrating LLMs with structured knowledge (KGs) in recommendation scenarios.

Beyond category-level transitions and brand loyalty, our framework is designed to be extensible to finer-grained item attributes—such as material, style, and complementary product relationships—as well as broadly applicable across domains beyond fashion (e.g., electronics, food, books). This generalizability is a deliberate design principle, not a secondary consideration.

4.2 Future Extensions

- Multi-modal intent inference: Incorporate product images and review text for richer intent modeling.
- Real-time adaptation: Extend the framework to support streaming purchase events for online recommendation.
- Cross-domain transfer: Apply the dynamic KG framework to other domains (books, electronics, food) where post-purchase category transitions are prevalent.
- Temporal modeling: Incorporate time-decay factors into brand loyalty and category transition patterns.

4.3 Relevance to Career Development

This project integrates multiple cutting-edge technologies—Knowledge Graphs, Graph Neural Networks, and Large Language Models—providing hands-on experience with the most in-demand skills in the AI/ML industry. The implementation-focused approach (using RecBole, building data pipelines, engineering LLM prompts) directly translates to industry-relevant engineering expertise, while the research component strengthens academic and analytical capabilities.

5. References

Courses & Lectures

1. Stanford University — CS224W: Machine Learning with Graphs.
2. Andrew Ng — Lecture 16: Recommender Systems.

Key Papers

3. Rajput, S., et al. (2023). *TIGER: Recommender Systems with Generative Retrieval*. NeurIPS 2023.
4. Lee, S., et al. (2025). *GRAM: Generative Recommendation via Semantic-aware Multi-granular Late Fusion*. ACL 2025.
5. Wang, S., et al. (2025). *K-RagRec: Knowledge Graph Retrieval-Augmented Generation for LLM-based Recommendation*. ACL 2025.
6. Chen, J., et al. (2025). *CORONA: A Coarse-to-Fine Framework for Graph-based Recommendation with Large Language Models*. SIGIR 2025.
7. Cai, R., et al. (2025). *Boosting Knowledge Graph-based Recommendations through Confidence-Aware Augmentation with Large Language Models*. arXiv preprint.
8. Balloccu, G., et al. (2023). *Faithful Path Language Modeling for Explainable Recommendation over Knowledge Graph*. arXiv preprint.

Datasets

9. Ni, J., Li, J., & McAuley, J. (2019). *Justifying Recommendations using Distantly-Labeled Reviews and Fine-Grained Aspects*. EMNLP 2019. (Amazon Review Data 2018)
10. Zhao, W. X., et al. (2021). *RecBole: Towards a Unified, Comprehensive and Efficient Framework for Recommendation Algorithms*. CIKM 2021.